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Extremely Low-Volume Burpee Interval Training Equivalent to 8 Minutes Per Session Improves Vertical Jump Compared with Sprint Interval Training in Real-World Circumstances

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¹Department of Physical Education and Health, Higher Institute of Physical Education, University of the Republic, Montevideo, Uruguay; ²Aquatic Sports Research Group, School of Physical Education, Physiotherapy and Dance, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil; and ³Department of Kinesiology, California State University San Marcos, San Marcos, California

Abstract

Pérez-Ifrán, P, Magallanes, CA, de S. Castro, FA, Astorino, TA, and Benítez-Flores, S. Extremely low-volume burpee interval training equivalent to 8 minutes per session improves vertical jump compared with sprint interval training in real-world circumstances. *J Strength Cond Res* XX(X): 000–000, 2023—The aim of this study was to compare the cardiometabolic and physical effects of 2 time-matched high-intensity programs in a real-world environment. Forty-three active and healthy adults (sex = 31 men and 12 women; age = 27 ± 5 years; peak heart rate [HR_{peak}] = 190.7 ± 10.6 beat·min⁻¹) were randomized to 2 very low-volume protocols (~8 minutes): sprint interval training (SIT) ($n = 15$), burpee interval training (BIT) ($n = 15$), and control (CON) ($n = 13$). Subjects in SIT and BIT performed 5 days of 10×4 second “all-out” efforts with 30 seconds of recovery. Body composition, blood pressure, countermovement jump (CMJ), 10-m sprint, shuttle run test (SRT), autonomic modulation, self-efficacy, and intention were evaluated before and after training. Sprint interval training elicited a higher % HR_{peak} , energy expenditure, rating of perceived exertion category ratio 10 scale, and feeling scale than BIT ($p < 0.05$). SRT_{distance} was significantly improved in SIT ($p = 0.03$, $d = 0.62$), whereas CMJ height was significantly enhanced in BIT ($p = 0.0014$, $d = 0.72$). Self-efficacy progressively worsened for SIT than for BIT as sessions increased, and significant differences were found in $5 \times$ a week frequency between protocols ($p = 0.040$, $d = 0.79$). No differences in intention to engage were detected between the regimens ($p > 0.05$). No changes were observed in body composition, blood pressure, 10-m sprint, SRT $\dot{V}O_{2max}$, or autonomic variables with training ($p > 0.05$). Results exhibit that extremely low-volume SIT improved running performance, whereas BIT increased the vertical jump.

Key Words: high-intensity interval training, high-intensity functional training, exercise prescription, cardiometabolic health, cardiovascular adaptations

Introduction

The prevalence of physical inactivity is a critical public health problem because currently, 7.2 and 7.6% of all-cause and cardiovascular deaths, respectively, are attributable to sedentary behavior (20). Moreover, physical inactivity dramatically reduces the quality of life and promotes disability and premature death because of its association with several noncommunicable diseases, including type 2 diabetes, hypertension, coronary heart disease, and several cancers (20). Paradoxically, obese individuals who are physically active and metabolically healthy (i.e., they have a high aerobic fitness i.e., maximum oxygen consumption [$\dot{V}O_{2max}$]) have a 30–50% lower risk of mortality from cardiovascular disease than obese adults with lower $\dot{V}O_{2max}$ (24). Maximum oxygen consumption is a significant predictor of the risk of all-cause mortality because an improvement equal to 3.5 ml·kg⁻¹·min⁻¹ is associated with a 13 and 15% lower risk of all-cause mortality and cardiovascular events, respectively (21). In addition, a slow heart rate recovery (HRR) after exercise is

established as an independent predictor of mortality (10), and heart rate variability (HRV) is inversely associated with all-cause mortality (22). Therefore, high $\dot{V}O_{2max}$ and high HR variability may mitigate the deleterious health consequences of sedentary lifestyle.

Current physical activity (PA) guidelines recommend 150–300 min wk⁻¹ of moderate-intensity continuous training or 75–150 min wk⁻¹ of vigorous-intensity PA to improve health status (8). However, these recommendations are not met by most adults as a large portion of the population is insufficiently active (8). Frequently, individuals report not having sufficient time to engage in PA (45), and thus, many clinicians have called for implementing more time-efficient PA programs. It is well documented that low-volume interval training promotes various physiological adaptations, including clinical (i.e., glycemic control, redox status, blood pressure, cardiac output, etc.) (2,5,40,46) and physical (i.e., $\dot{V}O_{2max}$, time trial performance, peak power output, etc.) (2,5,6,46) parameters in healthy and unhealthy adults. Interval training can be implemented using repeated submaximal efforts (high-intensity interval training [HIIT] at ≥ 80 of maximal HR) or “all-out” efforts (sprint interval training [SIT] at a power output $\geq 100\%$ of the workload associated with $\dot{V}O_{2max}$) (40).

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Sprint interval training has been identified as a very robust method to improve outcomes related to aerobic and anaerobic metabolism (41). Nevertheless, Wingate-based SIT is impractical for nonathletic adults because it elicits severe fatigue, hyperventilation, and nausea. Nevertheless, low-volume SIT characterized by a short-interval duration (i.e., 5 seconds) and fewer number of intervals have been proposed to reduce neuromuscular fatigue and enhance contribution of oxidative metabolism while improving subject tolerance (4). Despite the attenuated volume, short sprints have been shown to induce physiological adaptations, including significant improvements in $\dot{V}O_{2\max}$ and peak power output (6), in healthy adults and athletes. In addition, low-volume SIT has been shown to be effective in diabetic, prehypertensive, and obese patients in reducing the risk of cardiometabolic disease (46). Although experimental data clearly support the efficacy of low-volume SIT, the effectiveness of this approach under “real-world” conditions is controversial (46). Therefore, new studies are needed to investigate the effects of low-volume SIT with short bouts in real-world situations and potentially elucidate the best dose-response relationship to elicit physiological adaptations.

High-intensity functional training (HIFT) is a popular alternative to laboratory-based interval training because it does not require equipment and can be performed in a small space. These exercises are performed with body mass (i.e., burpees, mountain climbers, jumping jacks, etc.) and can be implemented in public environments such as parks, beaches, or sports courts. Chronic HIFT significantly enhances neuromuscular and cardiorespiratory function in clinical and healthy populations (39). For example, extremely low-volume (~8 minutes) HIFT significantly increases $\dot{V}O_{2\max}$ and time to exhaustion (29,38), muscle endurance, and power (29,38) and promotes beneficial changes in body fat and lean mass (25) in healthy adults. However, a recent meta-analysis summarizing 19 studies showed no changes in cardiometabolic variables (i.e., blood pressure, fasting glucose, or lipid profile) after 4–20 weeks of HIFT (39). Furthermore, HIFT exhibited lower increases in cardiorespiratory fitness yet greater changes in neuromuscular responses than traditional aerobic exercise (39). Nevertheless, to our knowledge, no study has compared chronic responses between HIFT vs. SIT, and only 2 studies compared the acute response to each regimen (3,14). In addition, the effects of HIFT, including short “all-out” bouts on cardiorespiratory fitness and neuromuscular performance, have yet to be explored.

The purpose of this study was to compare cardiometabolic and neuromuscular adaptations to 3 weeks of field-based HIFT and SIT with identical training volume. Developing exercise protocols in real-world scenarios is crucial to improve PA adherence and bridge the gap between laboratory and field exercise prescription. We hypothesize that very low-volume HIFT will attenuate cardiorespiratory adaptations but promote greater neuromuscular effects than SIT.

Methods

Experimental Approach to the Problem

To test the study’s hypothesis, we adopted a randomized controlled design including 2 experimental groups and a nontraining control group (CON). All groups performed 2 testing sessions (pre and post), and the experimental groups completed 5 very low-volume training sessions (~8 minutes) during a 3-week period. Previously, it was shown that 6 low-volume sessions of short (i.e., 5 seconds)

SIT improve cardiometabolic outcomes in young adults (5), although any effect of SIT having fewer sessions is unknown. Each training session was separated by 48–72 hours of recovery. This design was developed with the aim of being ecologically valid, using low-cost tools that are easy to use. Training was performed on an indoor basketball court with constant environmental conditions. All procedures were performed by the same evaluators, and to avoid effects of the circadian cycle, all sessions were held at the same time of day (8:00–12:00 PM). The incidental PA and dietary intake were monitored throughout the experiment.

Subjects

Initially, 87 individuals contacted the primary investigator after a publicity campaign at the University of the Republic, and 55 were recruited to participate in this study. From this sample, 50 volunteers met the inclusion criteria and were subsequently enrolled in the study. The inclusion criteria were as follows: (a) being moderately physically active (≥ 600 MET•min wk^{-1}) according to the International Physical Activity Questionnaire (IPAQ) Spanish short version (34), (b) not consuming any nutritional supplement, drugs, or tobacco products, (c) free of risk factors associated with cardiometabolic diseases, any musculoskeletal injury, and COVID diagnosis, (d) not using any beta-blockers drugs, (e) age between 18 and 40 years, and (f) if they were women, not be pregnant, or lactating. Subsequently, the subjects were randomly allocated into 3 groups and 43 completed the study: sex = 31 men and 12 women; age = 27 ± 5 years; height = 170.6 ± 8.9 cm; peak heart rate (HR_{peak}) = 190.7 ± 10.6 beat•min $^{-1}$; SIT; $n = 15$; burpee interval training (BIT); $n = 15$; and CON; $n = 13$. Seven subjects dropped out of the study for several reasons (Figure 1). The subjects were instructed to maintain their incidental lifestyle (work, hours of sleep, etc.) during the experiment, abstain from consuming alcohol for 48 hours before each session, and to avoid consumption of stimulants (mate, coffee, etc.) in the mornings. They were asked to wear loose clothing, be euhydrated, and not cycle or walk to the court before each test. Before their participation, the experimental procedures, benefits, and potential risks were explained to all subjects in written and verbal forms, and subsequently, they provided written informed consent. This study was approved by the University Ethics Committee of the Higher Institute of Physical Education, University of the Republic Uruguay (No. 2/2020 and date of final approval November 4, 2020) and was conducted following the principles stipulated in the Declaration of Helsinki (<https://sites.jamanetwork.com/research-ethics/index.html>).

Procedures

Testing Measures. Informed consent and various questionnaires were completed at the beginning of the first day, and subsequently, the subjects performed all assessments in the following sequence: (a) body composition, (b) blood pressure, (c) familiarization, (d) vertical jump, (e) 10-m sprint, and (f) shuttle run test (SRT, Figure 1). Exercise training included 5 sessions for each group, composed of 10 “all-out” bouts of 4-second duration with 30 seconds of recovery. This protocol was used in a previous acute study (3) and was included in a meta-analysis showing significant aerobic and anaerobic adaptations in response to SIT (6). The training intensity was monitored during each session using physiological and psychological outcomes. These tests were repeated at least 72 hours after the final session of training.

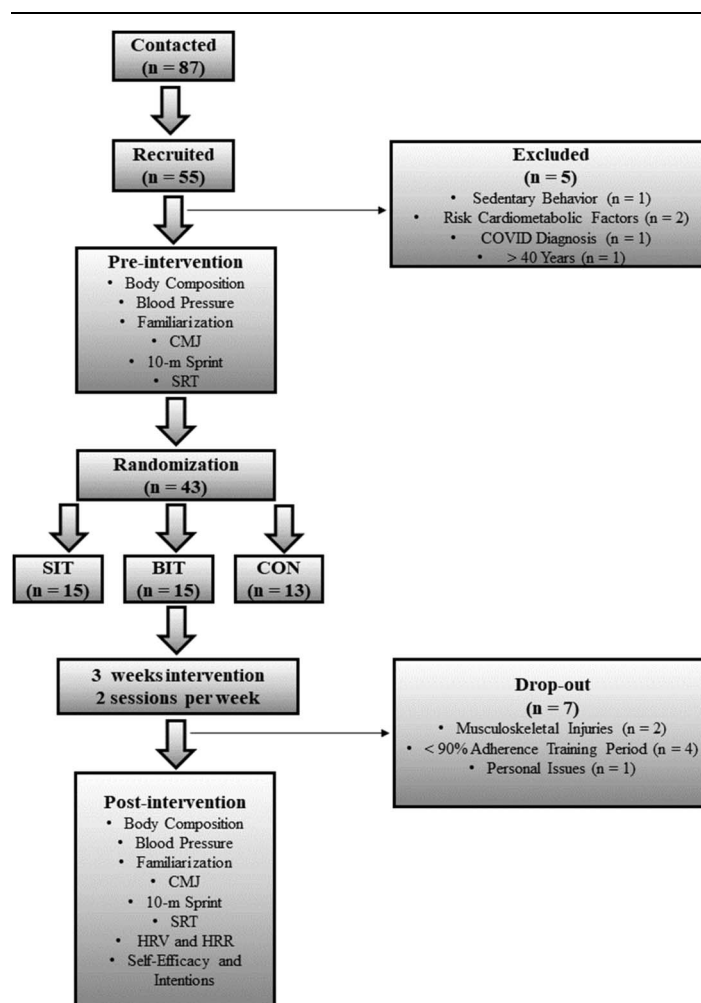


Figure 1. Study design. SIT = sprint interval training; BIT = burpees interval training; CON = control; CMJ = countermovement jump; SRT = shuttle run test; HRV = heart rate variability; HRR = heart rate recovery.

Body Composition We measured height (in centimeters) using a stadiometer (2096 PP; Toledo do Brazil, Sao Paulo, Brazil) and body mass (in kilograms), body mass index (BMI), muscle mass (%), fat mass (%), and visceral fat mass (%) using a digital body composition monitor (HBF 514C; OMRON, Kyoto, Japan). The measures were determined following previous procedures (32). To measure waist circumference (WC), the subjects assumed a relaxed standing position with the arms folded across the thorax, and WC was recorded at the level of the narrowest point between the lower costal (rib) border and the top of the iliac crest, at the end of normal expiration (32). Hip circumference (HC) was assessed at the level of the greatest posterior protuberance of the buttocks and perpendicular to the long axis of the trunk (32). Two measurements were taken with an anthropometric tape (SN-4010; Sanny Medical Starret, Sao Paulo, Brazil), using the mean value for analysis. The waist-to-hip ratio (WHR) was calculated by dividing WC and HC.

Blood Pressure Blood pressure was measured in a sitting position after 3 minutes of rest using a digital sphygmomanometer (CH-432; CITIZEN, Tokyo, Japan). This monitor has been shown to be reliable and valid to monitor blood pressure (11). The subject was seated with the arm slightly bent, the palm facing up, and the forearm supported nearly horizontal at the level of the heart. In addition, the

arm was free of tight clothing that could partially occlude blood flow (16). During the test, the legs were not crossed, and any form of isometric muscle action was to be avoided, such as pressing the legs down, hanging the feet off the ground, or sitting upright with the back unsupported. We positioned the cuff on the right arm with the lower margin about 2.5 cm above the antecubital space. Throughout the measurement, the arm was extended and supported at the level of the heart (16). Two measurements were performed following these procedures with a 3-minute rest interval between the assessments, and the mean value for diastolic blood pressure (DBP) and systolic blood pressure (SBP) was used for analysis.

Familiarization All subjects performed a 3-minute warm-up of running at a self-selected pace. Then, the subjects were familiarized with vertical jump and burpees. Initially, 2 repetitions of the countermovement jump (CMJ) were completed. The subjects were instructed to rest their hands on their hips while performing a downward movement to approximately 90° of knee flexion, followed by a maximal vertical jump, keeping their legs straight during the flight phase and landing at the same take-off point (26). Familiarization with the burpees was performed according to Gist et al. (14). The subjects stood with arms at the sides, then lowered into a squat position and placed the hands on the ground in front of the feet, then kicked the feet back and initiated a push-up, returned to a

squat position, and ended with a maximal jump with arms extended overhead. Approximately 5 minutes later, subjects were familiarized with SIT, which also served to provide practice for the 10-m sprint.

Countermovement Jump The CMJ is a validated and accessible method to determine lower-body power (26). The CMJ ($\text{CMJ}_{\text{height}}$) and absolute and relative body mass CMJ peak power ($\text{CMJ}_{\text{powerabs}}$, $\text{CMJ}_{\text{powerrel}}$) were recorded with PUSH band version 2.0 (PUSH Inc., Toronto, ON, Canada), which is valid and reliable for vertical jumps (30). The device was securely placed on the lower back with a waist belt. The average of 2 repetitions separated by 1 minute of passive recovery was analyzed.

10-m Sprint The 10-m sprint time was assessed, including 5-m split time. The subjects were instructed to run at maximum speed over a distance of 10 m, starting with the front foot 50 cm behind the starting line. Two repetitions were completed separated by passive recovery of 1 minute. Time to cover 10 m was recorded with *MySprint app* (Apple Inc., Cupertino, CA) that showed adequate validity and reliability for sprint time (35). *MySprint* was used with iPad (Apple Inc., Cupertino, CA) and was located 10 m away on a tripod, following the recommendations for measurements suggested by the creators of the APP (35). The mean of the 2 attempts was used for the analysis.

Shuttle Run Test The SRT is considered a valid approach to estimate $\dot{V}O_{2\text{max}}$ (43). The test consists of running for as long as possible between 2 lines separated by 20 m with a rhythm imposed by audio. The test starts with an initial speed equivalent to $8.5 \text{ km} \cdot \text{h}^{-1}$ and has increments equal to $0.5 \text{ km} \cdot \text{h}^{-1}$ every minute. The end of the test is determined when the 20-m distance cannot be covered in 2 consecutive efforts. To estimate $\dot{V}O_{2\text{max}}$, the equation proposed by Stickland et al. (43) was used: Men $\dot{V}O_{2\text{max}} = 2.75 \times [\text{last half-stage complete}] + 28.8$; Women $\dot{V}O_{2\text{max}} = 2.85 \times [\text{last half-stage complete}] + 25.1$. Also, HR_{peak} was recorded by employing chest straps through a telemetric system (Firstbeat Sports software version 4.7.3.1; Firstbeat Technologies Ltd., Jyväskylä, Finland). All subjects were verbally encouraged to exercise to exhaustion, and we used 2 criteria to classify the effort as maximum: (a) peak $\text{HR} \geq 90\%$ of the age-predicted maximum ($208 - [0.7 \times \text{age}]$); and (b) visible exhaustion. All subjects were verbally encouraged to exercise to exhaustion.

Dietary Intake During the training period, subjects tracked their dietary intake for 2 random days including 1 week day and 1 weekend day. A dietician instructed subjects how to complete the food diary and analyzed the information concerning total kcals, % carbohydrates, % proteins, and % fats. These data were analyzed using a custom-software MyFitnessPal (MyFitnessPal, Inc., San Francisco, CA) (44).

Incidental Physical Activity The incidental PA was assessed in all groups using IPAQ Spanish short version. This questionnaire was applied on the first and the last day of training to assess the PA done during the last week. It has adequate validity to quantify PA levels in young adults in contrast to accelerometry (34). The variables compared included (a) $\text{Kcal} \cdot \text{wk}^{-1}_{\text{total}}$; (b) $\text{Kcal} \cdot \text{wk}^{-1}_{\text{vigorous}}$; (c) $\text{Kcal} \cdot \text{wk}^{-1}_{\text{moderate}}$; and (d) $\text{Kcal} \cdot \text{wk}^{-1}_{\text{walking}}$.

Intervention Group Measures.

Interval Training Sessions Initially, subjects ran for 3 minutes at a self-selected pace. Total duration of each session was 5 minutes

10 seconds, and total exercise time was 40 seconds. Sprint interval training and BIT were equal in relation to the type of effort (“all-out”), type of stimulus (multijoint), and work-to-recovery ratio. Passive recovery was accomplished for both interventions, given that this type of rest was well tolerated across running-based SIT and BIT (28). During SIT, subjects were instructed to run as fast as they could, and after recovery, subjects were asked to run in the opposite direction. During BIT, subjects were encouraged to perform as many repetitions per interval as possible while maintaining correct technique. Burpees are a widely used exercise within HIFT routines; however, their unique longitudinal effects have not yet been revealed (39). Both regimens were monitored by a mobile application that provides audible alerts for each series (10×4 -second work and 30 seconds of recovery). Heart rate data were collected with a telemetric system (Firstbeat Sports software version 4.7.3.1; Firstbeat Technologies Ltd., Jyväskylä, Finland) that is valid and reliable (33), to describe internal load represented as the HR_{mean} and percentage of heart rate peak ($\%\text{HR}_{\text{peak}}$). Also, rating of perceived exertion category ratio 10 scale (CR-10 RPE) was used because it is designed to estimate the intensity of exercise, and it strongly correlated with HR (13). This instrument is graduated numerically from 0 to 10, with 0–2 ratings deemed easy effort, 3–6 ratings moderate to hard effort, and 7–10 ratings hard to maximum effort (13). This scale shows validity and good reliability for men and women (15). In addition, the feeling scale (FS), a psychometric tool that describes affective valence and emotional aspects of the exercise experience, focusing on the pleasure-displeasure dichotomy (17) was used to assess affective valence. This scale was validated in physically active individuals (1). The FS contains values from +5 to –5 (+5 equal “very good” and –5 represents “very bad”). Both scales were recorded immediately post-exercise. The physiological and psychological outcomes were recorded during the first, second, fourth, and fifth training sessions. Throughout the sessions, the researchers provided strong verbal encouragement for subjects to achieve maximum effort in each interval.

Heart Rate Variability and Heart Rate Recovery To assess the impact of the exercise protocols on autonomic balance, we analyzed the response of the HRV and HRR. Subjects remained in a supine position, which showed more reliability than other positions (7), and completed 2-minute records pre and post exercise in session first, second, fourth, fifth. They were requested to breathe normally and avoid any movements throughout data acquisition. To assess HRV, we analyzed only the second minute of recording (baseline and end for each session) because the first minute is considered a stabilization period during resting. This approach can assess autonomic function accurately in field environments and is easily applied during training routine (12). The variables selected were the R-R intervals and the root mean square of successive differences between R-R intervals (RMSSD), which is recognized as the strongest indicator of parasympathetic modulation (7). In addition, this parameter is not influenced by breathing frequency and can measure parasympathetic tone in a short period (7). Furthermore, the HRR was evaluated and defined as the difference between HR at the end of exercise (HR_{end}) and after 60 and 120 seconds of recovery ($\Delta\text{HR}_{\text{end-60s end}}$, $\Delta\text{HR}_{\text{end-120s end}}$). Also, we assessed the differences between baseline HR (HR_{bas}), and HR_{end} ($\Delta\text{HR}_{\text{bas-end}}$), 60 and 120 seconds ($\Delta\text{HR}_{\text{bas-60s end}}$, $\Delta\text{HR}_{\text{bas-120s end}}$).

Exercise Task Self-Efficacy and Intentions After the last training session, subjects’ confidence to repeat the exercise

protocol completed at different frequency per week (i.e., $1 \times$ a week to $5 \times$ a week) was assessed using a 5-item scale (19). Each question included the stem, “How confident are you that you can...”. The 5-items were as follows: (a) “perform one bout of exercise a week for the next 4 weeks that is just like the one you completed today?”; (b) “perform two bouts of exercise a week for the next 4 weeks that is just like the one you completed today?”; (c) “perform three bouts of exercise a week for the next 4 weeks that is just like the one you completed today?”; (d) “perform four bouts of exercise a week for the next 4 weeks that is just like the one you completed today?”; (e) “perform five bouts of exercise a week for the next 4 weeks that is just like the one you completed today?”. Responses were scored on a scale of 0% (not at all) to 100% (extremely confident) in 10% increments. Previously, the instrument demonstrated good internal consistency (α 's > 0.95) (19). Moreover, we asked the subjects regarding their ability to repeat the exercise using a 5-item measure, and the subjects' intentions to engage in the training regimens performed in the future at rates of 3 times or 5 times a week (intention $3 \times$ a week or intention $5 \times$ a week) over the next month (19). Specifically, subjects were asked “Please rate the extent to which you agree with the following statements”: (a) “I intend to engage in the type of exercise I performed today at least 3 times per week during the next month”, (b) “I intend to engage in the type of exercise I performed today at least 5 times per week during the next month.” The scores were registered using a Likert-type 7-point scale, ranging from 1 (very unlikely) to 7 (very probable).

Statistical Analysis

A priori power analysis through G*Power 3.1.9.7 (Dusseldorf University, Dusseldorf, Germany) was used to determine the required sample size to detect a change in $\dot{V}O_{2\max}$, considering the following input parameters: repeated-measures analysis of variance (ANOVA; within-between interaction), effect size $f = 0.25$; $\alpha = 0.05$; statistical power = 0.80, 3 groups, 4 measurements, correlation among repeated measures 0.4. The required sample size was 12 in each group, which was met in this study.

Descriptive statistics (mean $\pm$ SD and lower and upper limits—95%—of the confidence intervals) were calculated. Data normality was not tested because ANOVA is robust to normality infractions (36). Otherwise, when Student's t tests were applied, data normality was verified with Shapiro-Wilk test. To verify test reliability, intraclass correlation coefficients (ICC, 2-way mixed model for consistency analysis) were applied for WC, HC, SBP, DPB, CMJ_{powerabs}, CMJ_{powerrel}, and 5- and 10-m sprint time. To compare the baseline parameters between SIT, BIT, and CON, and dietary intake across SIT, BIT, and CON, 1-way ANOVA was applied. The Levene's test and the post hoc test (Bonferroni's if Levene's test result > 0.05 and Tamhane's if Levene's test result < 0.05) were executed.

To compare incidental PA, cardiometabolic, and physical parameters among SIT, BIT, and CON, a mixed, repeated-measures, 2×3 ANOVA (time and groups) was applied. To compare perceptual and heart rate response during training between SIT and BIT, a mixed, repeated-measures, 4×2 ANOVA (sessions and groups) was applied. To compare intentions between SIT and BIT, we used a mixed, repeated-measures, 2×2 ANOVA (intentions and groups). To compare self-efficacy between SIT and BIT groups, a mixed, repeated-measures, 5×2 ANOVA (bouts of exercise and groups) was applied. In all factorial ANOVAs, the main effects (with Bonferroni's post hoc test, when necessary) and interaction between factors were analyzed. When interactions were significant, splits were performed (1-way

ANOVA, with Bonferroni's post hoc test, when necessary, and Student's t test for dependent data). In all repeated and factorial ANOVAs, Mauchly's test identified the sphericity. When sphericity was not assumed, the Greenhouse-Geisser Epsilon correction was used. Training response, between SIT and BIT, was compared with independent t test.

To verify the effects sizes, partial eta-squared (η^2) and Cohen's d tests were performed. Interpretation of η^2 indicates small ($\eta^2 \geq 0.02$), medium ($\eta^2 \geq 0.13$), or large ($\eta^2 \geq 0.26$) effect sizes for a 2-way ANOVA and small ($\eta^2 \geq 0.01$), medium ($\eta^2 \geq 0.06$), or large ($\eta^2 \geq 0.14$) effect sizes for 1-way ANOVA. Interpretation of Cohen's d indicates 0–0.19 trivial, 0.2–0.59 small, 0.6–1.19 moderate, 1.2–1.99 large, 2.0–3.99 very large, and > 4.0 nearly perfect (9). The overall alpha level was set at $p < 0.05$. However, in the ANOVAs, the significance level was corrected for the number of comparisons, using the Bonferroni's correction factor. All analyses were performed with IBM SPSS version 23.0 (Armonk, NY).

Results

Reliability

Intraclass correlation coefficients were equal to 0.99, 0.86, 0.86, 0.86, 0.98, 0.96, 0.95, and 0.98, respectively, for WC, HC, SBP, DPB, CMJ_{powerabs}, CMJ_{powerrel}, and 5- and 10-m sprint time (all $p < 0.001$).

Dietary Intake and Incidental Physical Activity in All Groups

Total energy intake was equal to $1,252.0 \pm 414.5$ [816.9–1,687.0], $1,244.3 \pm 466.3$ [986.0–1,502.6], and $1,133.0 \pm 276.3$ [966.0–1,300.0] kcal for SIT, BIT, and CON, respectively ($F = 0.33$, $p = 0.71$, $\eta^2 = 0.02$). Carbohydrate intake was equal 48.2 ± 9.7 [42.8–53.6], 48.0 ± 6.5 [44.4–51.6], and 45.2 ± 8.4 [40.0–50.3] % for SIT, BIT, and CON, respectively ($F = 0.55$, $p = 0.57$, $\eta^2 = 0.002$). Fat intake did not differ across the 3 groups was 18.2 ± 4.0 [16.0–20.4], 17.8 ± 2.8 [16.3–19.4], and 17.6 ± 3.1 [15.7–19.6] % for SIT, BIT, and CON ($F = 0.087$, $p = 0.91$, $\eta^2 = 0.004$). Protein intake was not significantly different among groups: 32.0 ± 9.5 [26.7–37.3], 34.0 ± 5.9 [30.8–37.5], and 35.5 ± 9.9 [29.5–41.5] % for SIT, BIT, and CON ($F = 0.58$, $p = 0.56$, $\eta^2 = 0.02$).

No time $\times$ group interaction was detected for total PA kcals ($F = 0.33$, $p = 0.71$, $\eta^2 = 0.021$): pre-SIT $3,993.6 \pm 3,046.3$ [2,306.6–5,680.7] and post-SIT $2,680.1 \pm 2,065.9$ [512.0–4,848.1], pre-BIT $5,111.3 \pm 3,053.9$ [3,420.1–6,802.6] and post-BIT $4,897.6 \pm 3,067.4$ [3,199.0–6,596.3], pre-CON $2,946.3 \pm 1,675.2$ [1,934.0–3,958.7] and post-CON $3,508.0 \pm 2,000.6$ [2,299.0–4,716.9].

Compliance to Training in Intervention Groups

Four subjects dropped out (2 women and 2 men) for not attending $> 90\%$ of training sessions (Figure 1), and 1 dropped out for personal reasons. One man and 1 woman dropped out of the study because of musculoskeletal injuries (biceps femoris and rectus femoris tear, respectively) experienced during the SIT intervention. Fainting episodes, respiratory events, nausea, dizziness, and vomiting were not observed.

Training Response in Intervention Groups

During SIT, the subjects covered approximately 25-m each running bout. The training response was examined by considering

the average value across the 4 sessions for each regimen. The variables analyzed were as follows: (a) adherence; (b) %HR_{peak}; (c) energy expenditure; (d) CR-10 RPE; and (e) FS. Adherence % was not different between the groups (SIT = 93.3 ± 9.8 [87.9–98.3], BIT = 96.0 ± 8.3 [91.4–100.5], $t = -0.80$, $p = 0.42$, $d = 0.29$). Data show significant differences between the groups for %HR_{peak} (SIT = 83.2 ± 3.9 [81.1–85.3], BIT = 72.5 ± 9.2 [67.3–77.5], $t = 4.19$, $p < 0.001$, $d = 1.51$), energy expenditure (kcal) (SIT = 63.9 ± 13.4 [56.4–71.3], BIT = 47.5 ± 16.3 [38.5–56.9], $t = 2.99$, $p = 0.006$, $d = 1.09$), CR-10 RPE (SIT = 4.8 ± 1.8 [3.8–5.7], BIT = 2.6 ± 0.9 [2.0–3.0], $t = 4.27$, $p < 0.001$, $d = 1.55$), and FS (SIT = 2.7 ± 1.3 [2.0–3.4], BIT = 3.6 ± 1.0 [3.0–4.2], $t = -3.23$, $p = 0.002$, $d = 0.77$).

Changes in Cardiometabolic Parameters in All Groups

For body mass, BMI, muscle mass, fat mass, WC, HC, WHR, SBP, and DBP, no time × group interaction was detected ($p > 0.05$, $\eta^2 \leq 0.12$). Yet, a significant time × group interaction was noted for visceral fat mass ($p = 0.047$, $\eta^2 = 0.14$), and post hoc analysis showed that it was significantly reduced in response to BIT ($p = 0.05$, $d = 0.54$) compared with pretraining (Table 1).

Changes in Physical Parameters in all Groups

For CMJ_{powerabs}, CMJ_{powerrel}, 5-m sprint time, 10-m sprint time, SRT_{Velocity}, and SRT_{VO_{2max}}, results showed no time × group interaction ($p > 0.05$, $\eta^2 \leq 0.11$). However, there was a significant time × group interaction for CMJ_{height} ($p = 0.02$, $\eta^2 = 0.16$), and the post hoc analysis denoted higher CMJ_{height} in response to BIT ($p = 0.0014$, $d = 0.72$). Even though the improvement of CMJ height seems to be similar between BIT and CON, it should be noted that the coefficient of variation in CON (21.3%) was higher than in BIT (14.7%). Also, data exhibited a significant time × group interaction for SRT_{distance} ($p = 0.02$, $\eta^2 = 0.17$), and pairwise comparisons showed an increased SRT_{distance} in response to SIT ($p = 0.03$, $d = 0.62$) (Table 1).

Changes in Perceptual and Heart Rate Response in Intervention Groups

No time × group interaction for FS, CR-10 RPE, HR_{bas}, HR_{mean}, %HR_{peak}, Kcal_s, HR_{end}, HR_{60s end}, HR_{120s end}, RMSSD_{bas}, RMSSD_{end}, R-R_{bas}, or R-R_{end} was detected during the training sessions ($p > 0.05$, $\eta^2 \leq 0.05$; Table 2). In addition, no time × group interaction was noted for change scores for Δ HR_{end-60s end}, Δ HR_{end-120s end}, Δ HR_{bas-end}, Δ HR_{bas-60s end}, Δ HR_{bas-120s end}, Δ RMSSD_{bas-end}, and Δ R-R_{bas-end} during the training sessions ($p > 0.05$, $\eta^2 \leq 0.029$; Table 3).

Changes in Self-Efficacy and Intention Response in Intervention Groups

A significant frequency × group interaction was observed for self-efficacy after the training sessions ($F = 5.10$, $p = 0.032$, $\eta^2 = 0.15$). For SIT, the response was 93.3 ± 9.0 [88.3–98.3] for 1× a week, 84.0 ± 25.5 [69.8–98.1] for 2× a week, 73.3 ± 29.9 [56.7–98.1] for 3× a week, 58.0 ± 31.8 [40.3–75.6] for 4× a week, and 48.6 ± 29.9 [32.0–65.2] for 5× a week. For BIT, the response was 94.0 ± 8.2 [89.4–98.5] for 1× a week, 90.0 ± 15.1 [81.6–98.3] for 2× a week, 84.0 ± 19.9 [72.9–95.0] for 3× a week, 77.3 ± 23.4 [64.3–90.3] for 4× a week, and 71.3 ± 27.4

[56.1–86.5] for 5× a week. Significant intragroup differences were exhibited for SIT between 1× a week, 2× a week, and 3× a week vs. 4× a week, and 5× a week ($p < 0.05$, $d \leq 2.02$). Also, a significant intragroup difference was found for SIT between 4× a week vs. 5× a week ($p = 0.021$, $d = 0.30$). A significant intragroup difference was observed for BIT between 1× a week vs. 5× a week ($p = 0.034$, $d = 1.12$) and 2× a week vs. 4× a week and 5× a week ($p < 0.05$, $d \leq 0.84$). For the intergroup comparisons, the only difference was noted in 5× a week frequency, which being higher in BIT vs. SIT ($p = 0.040$, $d = 0.79$).

No intention × group interaction was detected for intention after the training sessions ($F = 0.43$, $p = 0.51$, $\eta^2 = 0.015$): intention 3× a week (SIT = 5.80 ± 1.26 [5.10–6.50], BIT = 5.27 ± 1.43 [4.47–6.06]), intention 5× a week (SIT = 4.07 ± 2.01 [2.95 to 5.18], BIT = 3.80 ± 1.61 [2.91 to 4.69]).

Discussion

This study is the first to compare cardiometabolic and physical adaptations to extremely low-volume HIIT (i.e., including only burpees, BIT) vs. SIT in real-world circumstances. During training, SIT elicited a greater physiological (i.e., %HR_{peak}) and psychological (i.e., CR-10 RPE, and FS) response vs. BIT. Results showed that only 5 sessions of BIT consisting of 120 seconds of exercise per week was sufficient to improve vertical jump height. Furthermore, very low-volume SIT significantly improved total distance covered in SRT. Nevertheless, no changes were observed in body composition, blood pressure, 10-m sprint, $\dot{V}O_{2max}$, or autonomic balance variables. Finally, the self-efficacy was progressively worse for SIT than for BIT as sessions increased. Our initial hypothesis was met given that HIIT improved neuromuscular performance without changes in cardiorespiratory fitness compared with SIT.

Prior research suggests that low-volume interval training improves glycemic control, antioxidant status, and $\dot{V}O_{2max}$ in healthy and unhealthy adults (2,5,6,39,40,46). A novel strategy to improve $\dot{V}O_{2max}$ is reducing the metabolic stress of traditional SIT using shorter efforts (6). For example, Hazell et al. (18) noted similar changes in power output, time trial performance, and $\dot{V}O_{2max}$ in response to 6 sessions of short (i.e., 10 seconds) vs. long bouts (i.e., 30 seconds) of SIT. Vollaard and Metcalfe (46) reported that 18 sessions of two 10–20 seconds sprints across a 10-minute session induce improvements in $\dot{V}O_{2max}$ (+12%) and insulin sensitivity in sedentary young adults. Benítez-Flores et al. (5) showed that 6–12 “all-out” 5-second bouts applied over 6 sessions, requiring only approximately 13 minutes of training, improved $\dot{V}O_{2max}$ (+7%) similar to subjects performing SIT or SIT combined with strength training in young healthy adults. In addition, Schaun et al. (38) demonstrated that 16 weeks of SIT consisting of eight 20 seconds efforts at 130% $\dot{V}O_{2max}$ similarly increased time to exhaustion and $\dot{V}O_{2max}$ vs. HIIT. Our study used 4-second bouts, and data show that SIT did not alter $\dot{V}O_{2max}$ yet significantly enhanced distance during SRT, representing a moderate effect ($d = 0.62$, +5%). This improvement is substantial given the high baseline $\dot{V}O_{2max}$ (i.e., >52 ml·kg⁻¹·min⁻¹) of our subjects. The findings corroborate those in a recent meta-analysis in which HIIT was found to attenuate increases in cardiorespiratory fitness than traditional aerobic exercise (39). These outcomes may be associated with the lower %HR_{peak} for BIT than for SIT observed in this study (Table 2) and a prior study (3). In addition, prior work shows peripheral adaptations in skeletal muscle related to the improved mitochondrial function, such as

Table 1
Cardiometabolic and physical parameters.*†

Variable	Pretime			Posttime			Group effect F , p , η^2	Time effect F , p , η^2	Time vs. Group F , p , η^2
	SIT	BIT	CON	SIT	BIT	CON			
Body mass (kg)	71.7 ± 11.7 (65.2–78.1)	70.3 ± 8.7 (65.4–75.1)	71.0 ± 16.2 (61.2–80.8)	71.3 ± 10.8 (65.3–77.3)	69.1 ± 8.7 (64.3–74.0)	70.9 ± 15.5 (61.5–80.3)	0.13; 0.87; 0.006	4.75; 0.035§; 0.10	1.49; 0.23; 0.06
BMI (kg·m ⁻²)	24.0 ± 1.9 (22.9–25.1)	24.5 ± 2.4 (23.1–25.8)	24.3 ± 3.0 (22.4–26.1)	23.9 ± 2.1 (23.0–24.9)	23.9 ± 2.1 (22.7–25.0)	24.2 ± 2.8 (22.5–25.9)	0.042; 0.95; 0.002	5.11; 0.029§; 0.11	2.58; 0.88; 0.11
Muscle mass (%)	36.1 ± 7.9 (31.7–40.4)	34.4 ± 6.2 (30.9–37.9)	33.8 ± 5.5 (30.4–37.1)	35.7 ± 8.2 (31.1–40.3)	35.3 ± 6.4 (31.7–38.9)	33.7 ± 5.7 (30.2–37.2)	0.77; 0.46; 0.035	0.38; 0.53; 0.01	2.23; 0.12; 0.10
Fat mass (%)	22.6 ± 7.2 (18.6–26.6)	27.2 ± 7.5 (23.1–31.4)	27.3 ± 5.9 (23.7–30.9)	22.5 ± 6.7 (18.8–26.3)	25.5 ± 7.5 (21.3–29.7)	27.4 ± 6.3 (23.6–31.2)	2.88; 0.067; 0.11	2.02; 0.16; 0.04	2.25; 0.11; 0.10
Visceral fat mass (%)	6.1 ± 1.6 (5.1–6.9)	6.6 ± 2.5 (5.1–8.0)	6.6 ± 3.0 (4.7–8.4)	6.0 ± 1.3 (5.2–6.7)	5.8 ± 2.0 (4.7–7.0)	6.6 ± 3.2 (4.7–8.6)	0.08; 0.92; 0.004	3.03; 0.08; 0.07	3.29; 0.047 ; 0.14
WC (cm)	76.7 ± 6.4 (73.1–80.3)	76.7 ± 5.7 (73.5–79.9)	76.4 ± 11.0 (69.7–83.1)	76.1 ± 5.6 (73.0–79.3)	76.6 ± 6.0 (73.2–79.9)	75.9 ± 10.9 (69.3–82.5)	0.058; 0.94; 0.003	3.63; 0.06; 0.083	0.42; 0.65; 0.02
HC (cm)	97.8 ± 5.1 (95.0–100.6)	99.5 ± 5.5 (96.4–102.5)	98.3 ± 8.0 (93.4–103.1)	98.0 ± 5.0 (95.2–100.8)	98.5 ± 4.7 (95.9–101.1)	97.9 ± 6.9 (93.7–102.1)	0.16; 0.85; 0.007	1.20; 0.27; 0.029	0.98; 0.38; 0.04
WHR	0.78 ± 0.05 (0.75–0.81)	0.77 ± 0.03 (0.74–0.79)	0.77 ± 0.07 (0.72–0.81)	0.77 ± 0.04 (0.75–0.80)	0.77 ± 0.04 (0.75–0.80)	0.77 ± 0.07 (0.72–0.81)	0.40; 0.66; 0.019	0.25; 0.61; 0.006	1.29; 0.28; 0.06
SBP (mm Hg)	124.9 ± 6.5 (121.3–128.5)	128.2 ± 10.1 (122.6–133.9)	135.8 ± 13.9 (127.4–144.2)	124.2 ± 7.3 (120.1–128.2)	126.8 ± 9.5 (121.5–132.1)	129.4 ± 11.3 (122.5–136.2)	1.43; 0.24; 0.063	5.96; 0.01§; 0.13	2.23; 0.12; 0.10
DPB (mm Hg)	74.7 ± 7.4 (70.6–78.8)	76.9 ± 6.1 (73.5–80.3)	83.9 ± 8.5 (78.7–89.0)	76.5 ± 8.3 (71.9–81.1)	75.4 ± 6.4 (71.8–79.0)	79.5 ± 9.4 (73.8–85.2)	2.44; 0.09; 0.10	1.72; 0.19; 0.04	2.91; 0.06; 0.12
CMJ _{height} (cm)	37.3 ± 7.1 (33.4–41.3)	33.1 ± 5.6 (30.0–36.3)	34.5 ± 5.7 (31.0–38.0)	36.5 ± 6.6 (32.8–40.2)	34.4 ± 6.2 (31.0–38.0)	36.0 ± 7.7 (31.3–40.7)	1.40; 0.25; 0.061	2.82; 0.10; 0.06	3.95; 0.02 ; 0.16
CMJ _{powerabs} (W)	3,179.5 ± 537.3 (2,882.0–3,477.1)	3,225.3 ± 605.1 (2,890.1–3,560.4)	2,999.7 ± 438.5 (2,734.7–3,264.7)	3,334.5 ± 591.8 (3,006.7–3,662.2)	3,279.3 ± 567.5 (2,965.0–3,593.5)	3,298.4 ± 552.4 (2,964.5–3,632.2)	0.56; 0.57; 0.026	10.8; 0.002§; 0.21	1.86; 0.16; 0.08
CMJ _{powerrel} (W·kg ⁻¹)	45.3 ± 9.7 (40.0–50.7)	46.2 ± 9.2 (41.1–51.4)	44.1 ± 11.0 (37.4–50.8)	47.8 ± 10.8 (41.8–53.8)	47.7 ± 8.4 (43.0–52.4)	48.7 ± 13.6 (40.4–57.0)	0.079; 0.92; 0.004	12.6; 0.001§; 0.24	1.28; 0.28; 0.06
5-m sprint time (s)	1.18 ± 0.08 (1.14–1.23)	1.23 ± 0.08 (1.19–1.28)	1.28 ± 0.12 (1.21–1.36)	1.16 ± 0.07 (1.12–1.20)	1.18 ± 0.08 (1.14–1.23)	1.27 ± 0.10 (1.21–1.33)	6.43; 0.004‡; 0.23	7.49; 0.009§; 0.15	1.24; 0.30; 0.05
10-m sprint time (s)	2.00 ± 0.15 (1.92–2.09)	2.08 ± 0.14 (2.00–2.16)	2.15 ± 0.18 (2.04–2.27)	1.97 ± 0.12 (1.91–2.04)	2.02 ± 0.15 (1.93–2.11)	2.14 ± 0.16 (2.03–2.24)	4.94; 0.012‡; 0.18	7.72; 0.008§; 0.16	0.80; 0.45; 0.03
SRT _{velocity} (km·h ⁻¹)	12.8 ± 1.09 (12.2–13.4)	12.6 ± 1.00 (12.0–13.9)	12.5 ± 0.97 (11.9–13.0)	13.0 ± 1.07 (12.4–13.6)	12.6 ± 1.09 (12.0–13.2)	12.3 ± 0.96 (11.8–12.9)	1.45; 0.24; 0.063	0.24; 0.62; 0.006	2.86; 0.08; 0.11
SRT _{VO_{2max}} (ml·kg ⁻¹ ·min ⁻¹)	54.5 ± 7.2 (50.5–58.5)	53.4 ± 6.6 (49.7–57.1)	52.6 ± 6.4 (48.7–56.7)	55.7 ± 7.0 (51.8–59.7)	53.4 ± 7.0 (49.5–57.4)	52.0 ± 6.4 (48.0–55.8)	1.22; 0.30; 0.054	0.24; 0.62; 0.006	2.69; 0.08; 0.11
SRT _{distance} (m)	1,628.0 ± 451.7 (1,377.8–1,878.1)	1,544 ± 417.9 (1,312.5–1,775.4)	1,504.6 ± 396.6 (1,264.9–1,744.3)	1,709.3 ± 461.0 (1,454.0–1,964.6)	1,553.3 ± 454.9 (1,301.3–1,805.2)	1,438.4 ± 368.8 (1,215.5–1,661.3)	1.37; 0.26; 0.16	0.15; 0.69; 0.004	4.18; 0.02 ; 0.17

*SIT = sprint interval training; BIT = burpees interval training; CON = control; BMI = body mass index; WC = waist circumference; HC = hip circumference; WHR = waist-to-hip ratio; SBP = systolic blood pressure; DBP = diastolic blood pressure; CMJ_{height} = countermovement jump height; CMJ_{powerabs} = countermovement jump peak power absolute; CMJ_{powerrel} = countermovement jump peak power relative; SRT_{velocity} = shuttle run test velocity; SRT_{VO_{2max}} = shuttle run test maximum oxygen consumption; SRT_{distance} = shuttle run test distance.

†Data are expressed as mean ± SD and limits of the mean confidence interval (95%).

‡Significant group effect.

§Significant time effect.

||Significant interactions between the factors; splits were performed: (a) visceral fat mass (%): analysis of group effects in the prettime and posttime, respectively: $F = 0.23$, $p = 0.23$, $\eta^2 = 0.01$ and $F = 0.50$, $p = 0.60$, $\eta^2 = 0.15$, respectively. Analysis of the effects of time in each group SIT, BIT, and CON, respectively: $t = 0.43$, $p = 0.67$, $d = 0.11$; $t = 2.120$, $p = 0.05$, $d = 0.54$; $t = -0.56$, $p = 0.58$, $d = 0.15$. (b) CMJ_{height} (cm): analysis of group effects in the prettime and posttime, respectively: $F = 1.77$, $p = 0.18$, $\eta^2 = 0.01$ and $F = 0.35$, $p = 0.60$, $\eta^2 = 0.15$. Analysis of the effects of time in each group SIT, BIT, and CON, respectively: $t = 1.15$, $p = 0.26$, $d = 0.19$; $t = -2.810$, $p = 0.0014$, $d = 0.72$; $t = -1.96$, $p = 0.07$, $d = 0.22$. (c) SRT_{distance} (m): analysis of group effects in the prettime and posttime, respectively: $F = 0.31$, $p = 0.73$, $\eta^2 = 0.01$ and $F = 1.38$, $p = 0.26$, $\eta^2 = 0.06$. Analysis of the effects of time in each group SIT, BIT, and CON, respectively: $t = -2.400$, $p = 0.03$, $d = 0.62$; $t = -0.26$, $p = 0.79$, $d = 0.06$; $t = 1.77$, $p = 0.10$, $d = 0.49$.

Bolded terms in footnote indicate significant differences.

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Table 2
Perceptual and heart rate response during interval trainings.*†

Variable	SIT				BIT				Group effect <i>F</i> , <i>p</i> , η^2	Session effect <i>F</i> , <i>p</i> , η^2	Session vs. Group effect <i>F</i> , <i>p</i> , η^2
	Session 1	Session 2	Session 3	Session 5	Session 1	Session 2	Session 3	Session 5			
FS	3.00 ± 1.85 (1.97–4.02)	2.40 ± 1.91 (1.33–3.46)	2.80 ± 1.65 (1.88–3.71)	2.80 ± 1.82 (1.79–3.80)	3.53 ± 1.64 (2.62–4.44)	3.46 ± 1.06 (2.87–4.05)	3.80 ± 1.01 (3.23–4.36)	3.80 ± 1.14 (3.16–4.34)	4.32; 0.04‡; 0.13	0.40; 0.55; 0.02	0.33; 0.79; 0.01
CR-10 RPE	4.8 ± 2.1 (3.6–6.0)	5.2 ± 2.3 (3.8–6.5)	4.9 ± 2.0 (3.8–6.0)	4.1 ± 1.8 (3.0–5.1)	2.8 ± 1.1 (2.1–3.4)	2.7 ± 0.8 (2.3–3.1)	2.4 ± 1.0 (1.8–3.0)	2.3 ± 1.0 (1.7–2.9)§	18.2; <0.001‡; 0.39	2.88; 0.053; 0.09	0.75; 0.52; 0.02
HR _{bas}	66.2 ± 12.7 (59.1–73.2)	64.8 ± 8.7 (59.9–69.6)	64.3 ± 10.4 (58.5–70.1)	60.0 ± 8.2 (56.1–65.2)	61.4 ± 10.4 (55.6–67.2)	58.0 ± 10.3 (52.3–63.8)	58.9 ± 11.0 (52.8–65.0)	60.1 ± 10.9 (54.0–66.2)	1.92; 0.17; 0.06	1.30; 0.27; 0.045	1.12; 0.34; 0.03
HR _{mean}	161.4 ± 15.4 (152.8–169.9)	161.0 ± 12.3 (154.2–167.8)	158.2 ± 12.9 (151.0–165.4)	158.6 ± 12.0 (151.9–165.3)	137.3 ± 18.9 (126.8–147.8)	136.7 ± 20.1 (125.6–147.8)	136.3 ± 20.6 (124.9–147.7)	139.6 ± 20.4 (128.3–150.9)	14.3; 0.001‡; 0.33	0.73; 0.53; 0.02	1.26; 0.29; 0.04
%HR _{peak}	84.0 ± 5.3 (81.0–87.0)	83.9 ± 4.4 (81.4–86.3)	82.4 ± 4.3 (80.0–84.8)	82.6 ± 4.2 (80.3–84.9)	72.4 ± 9.1 (67.3–77.5)	71.9 ± 10.1 (66.3–77.5)	71.8 ± 9.5 (66.5–77.1)	73.5 ± 9.5 (68.3–78.8)	17.5; <0.001‡; 0.38	0.75; 0.49; 0.02	1.29; 0.28; 0.04
Kcals	64.8 ± 14.5 (56.7–72.8)	64.0 ± 13.5 (56.5–71.6)	63.0 ± 13.7 (55.4–70.6)	63.7 ± 13.2 (56.3–71.0)	47.8 ± 16.9 (38.5–57.2)	47.0 ± 16.4 (37.9–56.2)	46.3 ± 17.0 (36.8–55.8)	48.9 ± 16.4 (39.7–58.0)	8.98; 0.006‡; 0.24	1.15; 0.33; 0.04	0.52; 0.66; 0.018
HR _{end}	172.1 ± 15.2 (163.6–180.4)	168.3 ± 11.3 (161.9–174.5)	166.1 ± 13.2 (158.7–173.3)	166.4 ± 12.6 (159.4–173.3)	152.9 ± 16.7 (143.6–162.1)	150.0 ± 18.7 (139.6–160.3)	149.0 ± 18.9 (138.5–159.5)	152.8 ± 18.9 (142.3–163.3)	9.44; 0.005‡; 0.25	3.63; 0.034§; 0.11	1.29; 0.28; 0.04
HR _{60s end}	125.2 ± 19.5 (114.3–136.0)	121 ± 16.1 (112.0–129.9)	117.9 ± 18.2 (107.7–127.9)	117.5 ± 18.1 (107.5–127.5)	100.8 ± 20.6 (89.3–112.2)	96.4 ± 20.9 (84.8–107.9)	94.2 ± 23.1 (81.4–107.0)	96.4 ± 26.7 (81.6–111.2)	11.1; 0.002‡; 0.28	3.65; 0.016§; 0.11	0.25; 0.85; 0.009
HR _{120s end}	105.6 ± 16.9 (96.1–115.0)	100.9 ± 12.5 (93.9–107.9)	98.0 ± 17.6 (88.3–107.8)	98.1 ± 16.5 (88.9–107.3)	87.2 ± 18.3 (77.0–97.3)	82.4 ± 18.0 (72.8–92.4)	80.6 ± 19.6 (69.9–91.7)	83.4 ± 21.8 (71.3–95.9)	7.79; 0.009‡; 0.21	5.77 0.001§; 0.170	0.50; 0.67; 0.018
RMSSD _{bas}	41.6 ± 20.1 (30.5–52.8)	40.8 ± 23.1 (28.0–53.6)	42.4 ± 16.4 (33.3–51.6)	53.3 ± 25.5 (39.2–67.4)	64.6 ± 32.6 (46.6–82.7)	68.3 ± 31.4 (50.9–85.7)	67.8 ± 26.5 (53.2–82.3)	72.6 ± 35.4§ (52.9–92.2)	7.61; 0.01‡; 0.21	2.49; 0.06; 0.08	0.39; 0.75; 0.01
RMSSD _{end}	10.4 ± 7.2 (6.4–14.4)	11.7 ± 9.1 (6.7–16.8)	12.0 ± 10.5 (6.2–17.9)	14.4 ± 14.5 (6.4–22.5)	25.2 ± 19.2 (14.5–35.9)	33.6 ± 25.5 (19.5–47.7)	35.8 ± 25.6 (21.7–50.0)	32.1 ± 25.3 (18.0–46.3)	9.72; 0.004‡; 0.25	3.26; 0.025§; 0.10	1.75; 0.16; 0.05
R-R _{bas}	932.7 ± 173.3 (836.7–1,028.7)	944.1 ± 147.8 (862.2–1,026.0)	946.1 ± 134.7 (871.5–1,020.6)	1,000.6 ± 125.7 (931.0–1,070.2)	994.4 ± 187.1 (890.8–1,098.0)	1,050.2 ± 202.6 (937.9–1,162.3)	1,037.2 ± 174.0 (940.8–1,133.6)	1,022.7 ± 185.7 (919.9–1,125.5)	1.76; 0.19; 0.05	1.25; 0.29; 0.04	1.05; 0.37; 0.03
R-R _{end}	426.8 ± 70.1 (388.0–465.6)	436.9 ± 56.9 (405.4–468.5)	449.9 ± 67.4 (412.6–487.3)	462.1 ± 65.6 (425.8–498.5)	555.7 ± 148.6 (473.4–638.0)	606.0 ± 175.9 (508.6–703.5)	626.5 ± 174.4 (529.9–723.2)	615.3 ± 207.0 (500.6–729.9)	11.56; 0.002‡; 0.29	5.70; 0.001§; 0.16	1.28; 0.28; 0.04

*SIT = sprint interval training; BIT = burpees interval training; FS = feeling scale; CR-10 RPE = rating of perceived exertion category ratio 10 scale; HR_{bas} = heart rate baseline; HR_{mean} = heart rate mean; %HR_{peak} = percentage of heart rate peak; HR_{end} = heart rate end; HR_{end-60s} = heart rate after 60 seconds of end; HR_{end 120s end} = heart rate after 120 seconds of end; RMSSD_{bas} = root mean square of successive differences between R-R intervals baseline; RMSSD_{end} = root mean square of successive differences between R-R intervals end; R-R_{bas} = R-R intervals baseline; R-R_{end} = R-R intervals end.
†Data are expressed as mean ± SD and limits of the mean confidence interval (95%).
‡Significant group effect.
§Significant session effect.

Table 3**Changes scores of heart rate during interval trainings.*†**

Variable	SIT				BIT				Group effect F ; p ; η^2	Session effect F ; p ; η^2	Session vs. Group effect F ; p ; η^2
	Session 1	Session 2	Session 3	Session 4	Session 1	Session 2	Session 3	Session 4			
$\Delta HR_{\text{end-60s}}$ end	46.8 ± 15.4 (38.3–55.4)	47.2 ± 11.2 (41.0–53.4)	48.2 ± 11.9 (41.5–54.8)	48.8 ± 12.2 (42.0–55.5)	52.1 ± 12.1 (45.4–58.8)	53.6 ± 12.8 (46.4–60.7)	54.8 ± 15.3 (46.2–63.3)	56.4 ± 14.4 (48.4–64.3)	2.23; 0.14; 0.074	1.07; 0.36; 0.03	0.12; 0.94; 0.005
$\Delta HR_{\text{end-120s}}$ end	66.4 ± 15.2 (58.0–74.9)	67.3 ± 7.5 (63.1–71.4)	68.0 ± 11.6 (61.5–74.4)	68.2 ± 11.8 (61.7–74.8)	65.7 ± 11.3 (59.4–72.0)	67.6 ± 10.9 (61.5–73.2)	68.2 ± 13.4 (60.7–75.6)	69.4 ± 11.5 (63.0–75.8)	1.02; 0.38; 0.035	0.003; 0.95; <0.001	0.11; 0.95; 0.004
$\Delta HR_{\text{bas-end}}$	105.8 ± 15.3 (97.3–114.3)	103.4 ± 7.9 (99.0–107.8)	101.7 ± 10.0 (96.1–107.2)	105.7 ± 12.5 (98.7–112.6)	91.4 ± 13.2 (84.1–98.7)	91.9 ± 13.9 (84.2–99.6)	90.1 ± 14.0 (82.3–97.8)	92.7 ± 13.4 (85.2–100.7)	9.62; 0.004‡; 0.25	1.21; 0.30; 0.042	0.26; 0.84; 0.010
$\Delta HR_{\text{bas-60s}}$ end	59.0 ± 16.0 (50.1–67.8)	56.2 ± 10.1 (50.7–61.8)	53.5 ± 16.1 (44.5–62.5)	56.8 ± 15.7 (48.1–65.5)	39.3 ± 15.3 (30.0–47.8)	38.3 ± 13.7 (30.7–45.9)	35.3 ± 15.1 (27.0–43.7)	36.3 ± 19.0 (25.7–46.8)	15.3; 0.001‡; 0.35	1.48; 0.22; 0.050	0.15; 0.92; 0.005
$\Delta HR_{\text{bas-120s}}$ end	39.4 ± 16.0 (30.5–48.2)	36.1 ± 10.2 (30.4–41.7)	33.7 ± 14.8 (25.5–41.9)	37.4 ± 14.8 (29.2–45.7)	25.7 ± 12.0 (19.0–32.4)	24.3 ± 10.4 (18.5–30.1)	21.9 ± 11.1 (15.7–28.1)	23.3 ± 13.2 (15.9–30.6)	9.51; 0.005‡; 0.25	2.11; 0.10; 0.07	0.21; 0.88; 0.008
$\Delta RMSSD_{\text{bas-end}}$ end	31.2 ± 21.3 (19.3–43.0)	29.0 ± 24.8 (15.2–42.8)	30.3 ± 19.3 (19.6–41.0)	38.9 ± 20.5 (27.5–50.2)	39.4 ± 30.4 (22.5–56.2)	34.6 ± 26.3 (20.0–49.2)	31.9 ± 23.8 (18.7–45.1)	40.5 ± 27.4 (25.2–55.7)	0.29; 0.59; 0.01	2.35; 0.07; 0.07	0.41; 0.74; 0.01
$\Delta R-R_{\text{bas-end}}$	505.8 ± 132.5 (432.4–579.2)	507.1 ± 113.9 (444.0–570.2)	496.1 ± 113.0 (433.5–558.7)	538.5 ± 110.7 (477.1–599.7)	438.7 ± 129.2 (367.1–510.2)	444.1 ± 114.2 (380.8–507.3)	410.6 ± 135.3 (335.6–485.5)	407.4 ± 104.6 (349.5–465.3)	7.12; 0.012‡; 0.20	0.36; 0.77; 0.013	0.84; 0.47; 0.029

*SIT = sprint interval training; BIT = burpees interval training; $\Delta HR_{\text{end-60s}}$ end = delta heart rate after 60 seconds of end; $\Delta HR_{\text{end-120s}}$ end = delta heart rate after 120 seconds of end; $\Delta HR_{\text{bas-60s}}$ end = delta heart rate after 60 seconds of end from baseline; $\Delta HR_{\text{bas-120s}}$ end = delta heart rate after 120 seconds of end from baseline; $\Delta RMSSD_{\text{bas-end}}$ end = delta root mean square of successive differences between R-R intervals of end from baseline; R-R_{bas-end} = R-R intervals of end from baseline.

†Data are expressed as mean ± SD and limits of the mean confidence interval (95%).

‡Significant group effect.

increased citrate synthase, cytochrome *c* oxidase subunit IV, and succinate dehydrogenase after 2 weeks of SIT (40). It is possible that some of these adaptations may explain gains in running performance in response to SIT, although further work is needed to confirm this result. Nevertheless, no significant increase in $\dot{V}O_{2\text{max}}$ was detected, possibly because of the very low dose of exercise per session (i.e., 40 seconds) and the total sessions performed (i.e., 5 sessions) compared with other studies showing significant increases in $\dot{V}O_{2\text{max}}$ with SIT (2,5,6,18,37,40). Despite these no significant findings, it is important to highlight that a gain of 1 ml·kg⁻¹·min⁻¹ in $\dot{V}O_{2\text{max}}$ was associated with a 9% relative risk reduction of all-cause mortality (23). Thus, our data could have substantial implications for the prevention of several diseases throughout adult life.

The importance of HR in detecting elevated risk of all-cause mortality has been identified (22). In fact, resting and exercise HR, HRR, and HRV are commonly used noninvasive field tools to evaluate adaptations in the regulation of the autonomic nervous system. After prolonged training, HRV and HRR fluctuate in accordance to training load and functional or nonfunctional overreaching (7). For example, moderate training loads generally promote an increase of vagal related HRV indices, whereas high training loads induce a decrease of vagal related HRV indices and slower HRR (7). Positive adaptations decrease the intrinsic rhythmicity of the heart through significant reductions in central sympathetic outflow as a consequence of enhancement in baroreflex control and chemoreflex sensitivity (31). Moreover, there are enhancements in peripheral blood flow with increase in nitric oxide synthesis and reduction in cytokines (31). However, we did not detect any change in indicators of autonomic status after 5 sessions of BIT or SIT (Tables 2 and 3). Regarding HRV, there are

divergent results concerning the efficacy of low-volume interval training (≥ 15 min session⁻¹). For example, 1 study using HIIT showed improvements in RMSSD and SDNN (42), whereas, another study using SIT did not (5). These equivocal data may be because of differences in training dose, as Songsorn et al. (42) implemented a longer dose of training vs. that of Benítez-Flores et al. (5) (6 vs. 2 weeks). Also, Songsorn et al. (42) recruited insufficiently active adults, whereas Benítez-Flores et al. (5) recruited moderately active adults. Our results also showed no effect of training on HRR. By contrast, Matsuo et al. (27) demonstrated enhanced HRR after 8 weeks of high-volume HIIT (3 × 3 minutes at 85% $\dot{V}O_{2\text{max}}$) in sedentary adults. The discrepant results may be explained by the different duration of recording, body position, training duration, type of exercise and training background.

Our results showed that BIT consisting of approximately 30 burpees per day significantly improved jump height. This is the first study exhibiting a moderate enhancement in muscle performance ($d = 0.72$, +4%; Table 1) in response to completing only burpees as part of HIIT. Nevertheless, there were no changes in absolute or relative peak power development during the jumps. Previously, Schaun et al. (37) reported an improvement in CMJ power output (+4%) and height (+6%) after 16 weeks of HIIT (8 minutes, 8 “all-out” × 20-second efforts) including 4 different exercises (burpees, mountain climbers, squat and thrusts with 3-kg dumbbells, and jumping jacks) in recreationally active adults. Evidence from a meta-analysis shows a significant effect of HIIT on 1 repetition maximum, longitudinal and vertical jump, and strength endurance (39). Therefore, HIIT may be a promising strategy for improving strength in active, non highly trained adults.

No exercise protocol is suitable for wide implementation unless it is well-tolerated by clientele. Our results showed no main effect or interaction for intention $3\times$ a week or $5\times$ a week for BIT or SIT. Previously, a greater intention to participate $5\times$ a week was noted for BIT vs. SIT (28), which does not support our data. Moreover, we observed a significant trend toward lower self-efficacy across sessions for both groups. The self-efficacy score was progressively worse for SIT than for BIT as sessions increased, and significant differences were found in $5\times$ a week frequency ($p = 0.040$, $d = 0.79$). Yet, the response was considerably positive for both protocols in self-efficacy and intention up to $3\times$ a week frequency (i.e., $\geq 73/100$ self-efficacy, $\geq 5/7$ intention). These findings may be associated with minimal changes in CR-10 RPE (≤ 5 , hard) and FS (≥ 3 , good) during training, although a worse response was detected for SIT vs. BIT.

This study has a few limitations. First, data can only be applied to young, active, nonobese adults. Second, we implemented a very low dose of SIT and HIFT, and different findings may occur from regimens characterized by a higher training volume. Third, neither muscular strength nor $\dot{V}O_{2\max}$ was measured directly in this study, and further work is needed to determine if these laboratory-based measures can be improved in response to low-volume BIT and SIT. Fourth, although HR is an ecological tool for assessing internal load, it is very sensitive to environmental changes.

Practical Applications

This study was designed to examine the implementation of extremely low-volume interval training in real-world circumstances. Our data in active adults show that 5 sessions of very low-volume running-based SIT enhances run distance, whereas BIT improves vertical jump height. Yet, no improvements were noted in body composition, blood pressure, $\dot{V}O_{2\max}$, or autonomic modulation. These results exhibit fitness-related benefits with very low dose of exercise equivalent to 40 s d^{-1} with good adherence, tolerance, self-efficacy, and intention to engage $3\times$ a week frequency. Thus, strength and conditioning professionals and enthusiasts should consider planning sessions with approximately 10 very short yet intense bouts to potentially mitigate sedentary behavior. However, strength and conditioning professionals have to apply running-based SIT with caution because it can cause musculoskeletal injury, and we suggest a preparatory training plan before starting “all-out” running SIT (e.g., submaximal sprint bouts) or completing other modalities (e.g., cycling or rowing). Another option is to design programs with HIFT because it is a modality with promising results in health indicators that has been established as one of the main trends in the fitness industry. Future studies should replicate these protocols in untrained, sedentary, or at-risk populations.

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Magallanes: conceived and designed the investigation, analyzed and interpreted the data, drafted the paper, and approved the final version submitted for publication. F. A. de S. Castro: analyzed and interpreted the data, drafted the paper, and approved the final version submitted for publication. T. A. Astorino: analyzed and interpreted the data, drafted the paper, and approved the final version submitted for publication. S. Benítez-Flores: conceived and designed the investigation, analyzed and interpreted the data, drafted the paper, and approved the final version submitted for publication. Institutional Ethics Committee Statement: The Ethics Committee from the Higher Institute of Physical Education, University of the Republic approved this study in accordance with the Declaration of Helsinki (November 4, 2020). Funding: This work was supported by the [CSIC, University of the Republic, Uruguay] under Grant [192, 2018].

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